

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int function(int n)
4  {
5      int count = 0;
6      int i, s;
7
8      i = 1;
9      count++;
10     s = 1;
11     count++;
12
13     while (1)
14     {
15         count++;
16         if (!(s <= n))
17             break;
18         i++;
19         count++;
20         s += i;
21         count++;
22     }
23
24     return count;
25 }
26
27 int main()
28 {
29     int n;
30     scanf("%d", &n);
31     printf("%d", function(n));
--
```

```
32 | return 0;  
33 | }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("");
                printf("");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, c = 0;
6      scanf("%d", &n);
7
8      if(n == 1)
9      {
10         c = 1;
11     }
12     else
13     {
14         c = 5 * n + 2;
15     }
16
17     printf("%d", c);
18
19     return 0;
20 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void Factor(int num)
4  {
5      int i, c = 0;
6      for (i = 1; i <= num; ++i)
7      {
8          c++;
9          c++;
10         if (num % i == 0)
11         {
12             c++;
13         }
14     }
15     c++;
16
17     printf("%d", c);
18 }
19
20 int main()
21 {
22     int n;
23     scanf("%d", &n);
24     Factor(n);
25     return 0;
26 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓

	Input	Expected	Got	
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, j, k, c = 0;
6
7      scanf("%d", &n);
8
9      c++;
10
11     for (i = n / 2; (c++, i < n); i++)
12     {
13         for (j = 1; (c++, j < n); j = 2 * j)
14         {
15             for (k = 1; (c++, k < n); k = k * 2)
16             {
17                 c++;
18             }
19         }
20     }
21
22     printf("%d", c);
23
24     return 0;
25 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, rev = 0, remainder, c = 0;
6
7      scanf("%d", &n);
8
9      c++;
10
11     while (n != 0)
12     {
13         c++;
14         remainder = n % 10;
15         c++;
16         rev = rev * 10 + remainder;
17         c++;
18         n /= 10;
19         c++;
20     }
21
22     c += 2;
23
24     printf("%d", c);
25
26     return 0;
27 }
```

	Input	Expected	Got	
✓	12	11	11	✓

	Input	Expected	Got	
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)